

1. The *DietInfo* class is the originator class. It contains 3 fields: *personName*, *days*, *weight*. The *DietInfo* class contains a private inner class called *DietInfoMemento*. The *save()* method of *DietInfo* creates and returns a *DietInfoMemento* object. The *restore()* method of *DietInfo* is used to restore the state of the *DietInfo*.

DietInfoCaretaker is the caretaker class that is used to store the state of a *DietInfo* object. The memento is stored as an object since *DietInfo.DietInfoMemento* is not visible to the caretaker. The caretaker's *saveState()* method saves the state of the *DietInfo* object. The caretaker's *restoreState()* method restores the state of the *DietInfo* object.

The *Test* class demonstrates the memento pattern. It creates a *DietInfoCaretaker* and then a *DietInfo* object. The *DietInfo* object's state is changed and displayed. At one point, the caretaker saves the state of the *DietInfo* object. After this, the *DietInfo* object's state is further changed and displayed. After this, the caretaker restores the state of the *DietInfo* object. Finally, verify this restoration by displaying the *DietInfo* object's state.

DietInfo

```
package experiment19_1;

class DietInfo {

    private String personName;

    private int days;

    private double weight;

    public DietInfo(String personName, int days, double
weight){

        this.personName = personName;

        this.days = days;
```

```
        this.weight = weight;
    }

    public void setPersonName(String personName) {
        this.personName = personName;
    }

    public void setDays(int days) {
        this.days = days;
    }

    public void setWeight(double weight) {
        this.weight = weight;
    }

    public DietInfoMemento save(){
        return new DietInfoMemento(personName, days,
weight);
    }

    public void restore(DietInfoMemento dietInfoMemento){
        this.personName = dietInfoMemento.personName;
        this.days = dietInfoMemento.days;
    }
}
```

```
        this.weight = dietInfoMemento.weight;
    }

    public String getPersonName() {
        return personName;
    }

    public int getDays() {
        return days;
    }

    public double getWeight() {
        return weight;
    }

    class DietInfoMemento{
        private String personName;
        private int days;
        private double weight;

        private DietInfoMemento(String personName, int
days, double weight){
```

```

        this.personName = personName;

        this.days = days;

        this.weight = weight;
    }

    private String getPersonName() {
        return personName;
    }

    private int getDays() {
        return days;
    }

    private double getWeight() {
        return weight;
    }
}

```

DietInfoCaretaker

```

package experiment19_1;

class DietInfoCaretaker {

```

```

    private DietInfo.DietInfoMemento memento;

    public void saveState(DietInfo dietInfo){
        memento = dietInfo.save();
    }

    public void restoreState(DietInfo dietInfo){
        dietInfo.restore(memento);
    }
}

```

Test

```

package experiment19_1;

public class Test {
    public static void main(String[] args) {
        DietInfoCaretaker dietInfoCaretaker = new
DietInfoCaretaker();

        DietInfo dietInfo = new DietInfo("John Doe", 30,
75.5);

        System.out.println("Initial state:");

        System.out.println("Person: " +
dietInfo.getPersonName() + ", Days: " + dietInfo.getDays() + ",
Weight: " + dietInfo.getWeight());
    }
}

```

```
dietInfoCaretaker.saveState(dietInfo);

dietInfo.setDays(25);

System.out.println("\nState after changes:");

System.out.println("Person: " +
dietInfo.getPersonName() + ", Days: " + dietInfo.getDays() + ",
Weight: " + dietInfo.getWeight());

dietInfoCaretaker.restoreState(dietInfo);

System.out.println("\nState after restoration:");

System.out.println("Person: " +
dietInfo.getPersonName() + ", Days: " + dietInfo.getDays() + ",
Weight: " + dietInfo.getWeight());

    }
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ IDEA 202
```

```
Initial state:
```

```
Person: John Doe, Days: 30, Weight: 75.5
```

```
State after changes:
```

```
Person: John Doe, Days: 25, Weight: 75.5
```

```
State after restoration:
```

```
Person: John Doe, Days: 30, Weight: 75.5
```

```
Process finished with exit code 0
```